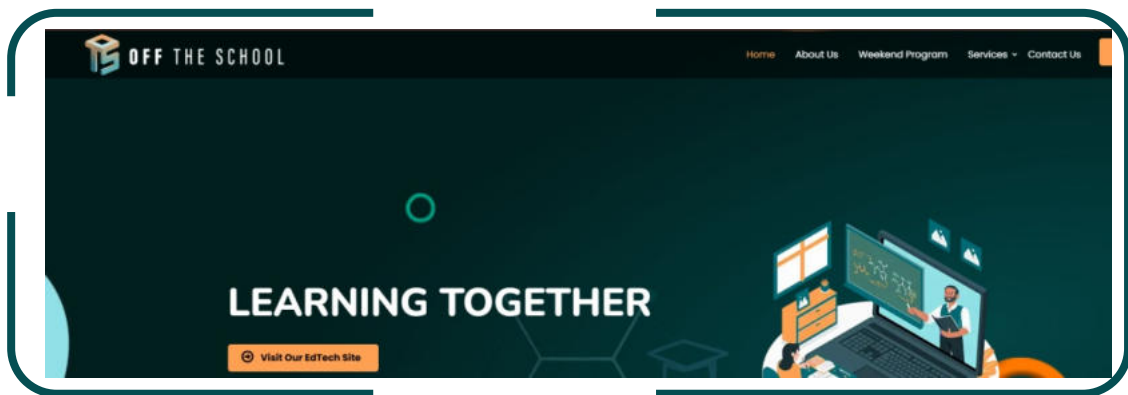


UNIT 4

INTRODUCTION TO WEB DEVELOPMENT – HTML BASICS



Learning Objectives

- Define the key concepts of HTML, including hypertext, markup, and language.
- Identify and explain the different versions of HTML.
- Understand the basic structure and syntax of HTML tags.
- Create simple web pages using HTML, incorporating both basic and for matting tags.
- Demonstrate the use of attributes in HTML tags.
- Recognize the difference between start and closing tags and explain the importance of correct tag syntax.
- Create a simple HTML webpage with elements like headings, lists, and formatted text.

What is Web Development?

Web development is the process of creating websites that you can view on the internet using a browser like Google Chrome or Firefox. It includes designing the layout, writing content, and using programming languages like **HTML**, **CSS**, and **JavaScript**.

Introduction to HTML

HTML stands for Hyper Text Markup Language. It is a formatting language used to define the appearance and contents of a web page. It allows us to organize text, graphics, audio, and video on a webpage.

Key Points:

- The word *Hypertext* refers to the text which acts as a link.
- The word *Markup* refers to the symbols that are used to define the structure of the text. The markup symbols tell the browser how to display the text and are often called tags.
- The word *Language* refers to the syntax that is similar to any other language.

HTML was created by **Tim Berners-Lee** at CERN.

HTML Versions:

The following table shows the various versions of HTML and their release years:

Version	Year
HTML 1.0	1991
HTML 2.0	1995
HTML 3.2	1997
HTML4.0	1999
XHTML	2000
HTML 5	2012
HTML 5.1	2016
HTML 5.2	2019

Note:

HTML 5 introduced several new features, including improved support for multimedia, APIs, and new semantic elements. HTML 5.1 and HTML 5.2 focused on updates and bug fixes to HTML5, and are considered part of the ongoing HTML5 standard.

HTML Tags

Tag is a command tells the web browser how to display the text, audio, graphics or video on a web page.

Key Points:

A **tag** is a command that tells the web browser how to display text, audio, graphics, or video on a web page.

- Tags are enclosed in angle brackets (< >).
- They start with a less-than (<) character and end with a greater-than (>) character.
- Most HTML tags occur in pairs: a start tag and a closing tag.
- The **start tag** is simply the tag name enclosed in angle brackets, while the **closing tag** includes a forward slash (/) before the tag name.
- Some tags are self-closing and don't require a closing tag.
- Tags are not case-sensitive, but it is best practice to use lowercase.

Basic Tags

Following table shows the basic HTML tags that define the web page:

Tag	Description
<html> </html>	Specifies the document as a web page.
<head> </head>	Specifies the descriptive information about the web document
<title> </title>	Specifies the title of the web page.
<body> </body>	Specifies the body of the web document.

Example code:

The following code shows how to use basic tags.

```
<head> Heading goes here...</head>
<title> Title goes here...</title>
<body> Body goes here...</body>
</html>
```

Formatting Tags:

The following table shows few HTML tags used for formatting the text:

Tag	Description
<code> </code>	Specifies the text as bold. e.g., this is bold text.
<code><i> </i></code>	The content of italic tag is displayed in italic. e.g., <i>Italic text</i>
<code><sub> </sub></code>	Specifies the subscripted text. e.g., X_1
<code><sup> </sup></code>	Defines the superscripted text. e.g., X^2

HTML Lists

- HTML lists allow web developers to group a set of related items in lists.
- HTML offers three ways for specifying lists of information. All lists must contain one or more list elements.

Types Of Lists:

Ordered List:

- Ordered list also called as Numbered list, is used to present a numbered list of item in the order of importance or the item (paragraph) is marked with a number.

- An ordered list must begin with the `<ol>` followed by an `<li>` list item tag.

```
<ol>

<li>Fruits</li>

<li>Vegetables</li>

<li>Milk</li>

</ol>
```

Unordered List:

- Unordered List also called as bulleted list, used to present list of items marked with bullets. An unordered list starts with in `<UL>` followed by `<LI>` (List Item) tag. Use of `<UL>` is very similar to `<OL>` (ordered list).

```
<ul>

<li>Fruits</li>

<li>Vegetables</li>

<li>Milk</li>

</ul>
```

Working with HTML using Notepad

You can create and edit HTML files using Notepad, which is a simple text editor available on most computers. Here's how you can get started:

1. Open Notepad

- On Windows, click on the *Start menu*, type "Notepad", and press Enter.

2. Create an HTML file

- In Notepad, start typing your HTML code. For example:

```
<html>

  <head>
    <title>My First Web Page</title>
  </head>
  <body>
    <h1>Welcome to My Web Page!</h1>
    <p>This is a paragraph of text.</p>
  </body>
</html>
```

3. Save the file

- After writing your HTML code, save the file as .html.
- Click File > Save As.
- In the Save as type dropdown, choose All Files.
- Name your file home.html and click Save.

4. View the file in a browser

- Locate the file you just saved (e.g., home.html) and double-click it. It will open in your default web browser where you can view your web page.

4. Editing the file

- To make changes, right-click the file and choose *Edit*. This will reopen the file in Notepad, where you can make changes to the HTML code.

After saving, refresh the page in the browser to see the changes.

This process allows you to create and edit HTML files easily with a simple text editor like Notepad. This will help you get hands-on experience working with HTML and creating basic webpages.

Project Work

Create a web page in HTML with following features

- The file name should be home.html.
- The title of the page should be “Unit 2”.
- An unordered list with two items: System software & Application software.
- List the types of System software and Application software.

EXERCISES

01: Choose the correct answer.

1.What does HTML stand for?

- a) Hyperlink Text Markup Language
- b) Hypertext Markup Language
- c) Hyper Transfer Markup Language
- d) Hypertext Modifying Language

2.Which version of HTML was introduced in 2012?

- a) HTML 5
- b) HTML 4.0
- c) HTML 3.2
- d) XHTML

3.Which of the following tags is used to specify the title of the webpage?

- a) <body>
- b) <html>
- c) <title>
- d) <head>

4.Which tag is used to make text bold in HTML?

- a)
- b) <i>
- c) <sub>
- d) <sup>

5. In which year was HTML 4.0 introduced?

- a) 1995
- b) 1999
- c) 2012
- d) 2000

6. Which of the following HTML tags is used for the body of a webpage?

- a) <body>
- b) <html>
- c) <head>
- d) <footer>

7. HTML tags are _____.

- a) case-sensitive
- b) only used for formatting text
- c) not case-sensitive
- d) always written in uppercase

8. Which of the following tags is used to define a webpage as a web document?

- a) <body>
- b) <html>
- c) <head>
- d) <footer>

9. What does the <i> tag do in HTML?

- a) Italicizes text
- b) Adds a heading
- c) Makes text bold
- d) Adds a paragraph

10. Which HTML tag is used to define subscripted text?

- a) <sup>
 - b) <sub>
 - c) <i>
 - d)
-

02: Fill in the blanks:

1. HTML stands for _____ Markup Language.
2. The _____ tag is used to define the head section of a web page.
3. The _____ tag is used for defining bold text.
4. HTML 5 was introduced in the year _____.
5. The _____ tag is used to define the body section of a webpage.
6. The <_____> tag is used to specify the title of the webpage that appears in the browser.
7. HTML tags are usually enclosed in _____ brackets.
8. The _____ tag makes text appear in italics.
9. In HTML, the <html> tag defines the _____ of the webpage.
10. The <sub> tag is used to define _____ text.

03: Select True or False for each statement.

	TRUE	FALSE
1. The <i> tag is used to define bold text.	☐	☐
2. HTML 5 was released in 1999.	☐	☐
3. HTML tags are case-sensitive.	☐	☐
4. The <title> tag defines the title that appears in the browser tab.	☐	☐
5. The <body> tag is used to define the head of the webpage.	☐	☐
6. HTML tags always need to be in lowercase.	☐	☐
7. The <html> tag is used to specify the document as a web page.	☐	☐
8. The <head> section contains the main content of the webpage.	☐	☐
9. The <sub> tag is used to make text italic.	☐	☐
10. HTML 4.0 was introduced in 1997.	☐	☐

04: Answer the following questions briefly:

1. What does HTML stand for?
2. Explain the difference between the <head> and <body> tags in HTML.
3. What is HTML 5 and what are some features it introduced?
4. Name three formatting tags used in HTML and explain their functions.
5. How are start and closing tags different in HTML? Provide an example.
6. Why is it important to use proper tag syntax in HTML?

05: Provide detailed answers of the following questions:

1. Write the correct HTML code to display:

Welcome to Grade 8!

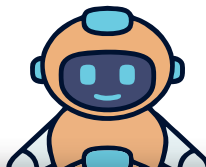
2. What is the first step to start writing HTML using Notepad?
3. Write down the basic structure of an HTML document.
4. Write the steps to save an HTML file using Notepad.

Lab Activity 1: Create a Simple HTML Page



- Using Notepad, create an HTML page with the following:
 - Title: "Unit 2"
 - An unordered list with two items: "System software" and "Application software"
 - Add sublists under each item, listing types of System software and Application software.
 - Save the file as home.html and open it in a web browser to check the display.

Lab Activity 2: Apply Formatting Tags



- Create an HTML document with the following:
 - Add a heading using `<h1>` tag.
 - Use `<b>` for bold text, `<i>` for italic text, `<sub>` for subscript, and `<sup>` for superscript.
 - Display the text "H₂O" and apply both subscript and superscript formatting to it.
 - Save and view the file in a browser.